RESEARCH PROPOSAL — DEVELOPMENT ECONOMICS / APPLIED MICROECONOMICS

Financial Literacy and Economic Empowerment among Female Sex Workers in Colombia

Evidence from a Randomized Controlled Trial

Intervention Instrument:

Ruta Financiera: El Legado

(AI-powered gamified mobile financial education platform)

Submitted for Review · June 2026

Keywords: financial literacy, sex work, RCT, behavioral economics, Colombia, digital interventions, economic empowerment

JEL Codes: D14, I25, J16, O12, C93

Abstract

Female sex workers (FSWs) in Colombia face acute and compounding financial vulnerability: irregular and cash-based income streams, limited access to formal financial services, high exposure to coercive financial relationships, and near-total exclusion from mainstream financial education programs. Despite a rich literature on financial literacy interventions and their effects on savings, credit use, and consumption smoothing, no experimental study has targeted this population in the Latin American context.

This proposal presents the design of a randomized controlled trial (RCT) that evaluates the causal impact of a gamified, AI-powered financial education platform — *Ruta Financiera: El Legado* — on the financial knowledge, attitudes, behaviors, and economic outcomes of FSWs across three major Colombian cities: Bogotá, Medellín, and Cali. The platform, already developed and piloted, delivers five sequential learning modules covering personal budgeting, credit and debt management, basic investment, entrepreneurship, and formal business registration, using conversational AI tutors, interactive exercises, and narrative gamification adapted to Colombian cultural referents.

We implement a two-arm individually randomized design, stratifying by city and baseline financial literacy score, targeting $N = 900$ participants (450 treatment, 450 control) recruited through community organizations, harm-reduction networks, and peer outreach using respondent-driven sampling (RDS). Primary outcomes are financial literacy scores and formal savings rates at six and twelve months post-randomization. Secondary outcomes include household consumption volatility, credit use, income diversification, and an index of economic autonomy. We complement the experimental design with OLS estimators, heterogeneous treatment effect analysis by vulnerability subgroup, and a mediation analysis framework to unpack behavioral mechanisms.

Contributions: (i) first experimental estimates of a digital financial education intervention targeting FSWs in a middle-income country; (ii) a test of whether gamified, narrative-based digital content generates durable behavioral change in a population with irregular schedules and limited prior exposure to formal finance; (iii) design principles for economic inclusion programs reaching stigmatized and hard-to-survey populations.

Contents

1. Introduction

Financial exclusion is both a cause and a consequence of poverty. Households without access to formal savings instruments cannot smooth consumption against income shocks; those without credit access cannot invest in productive opportunities; those without insurance face catastrophic losses from health or income disruptions. The causal links between financial literacy — defined as the knowledge, skills, and confidence to make effective financial decisions — and these outcomes have received substantial attention in the economics literature over the past two decades (Duflo and Banerjee, 2007; Karlan and Zinman, 2008; Lusardi and Mitchell, 2014).

Yet this literature has a critical blind spot: it almost exclusively targets populations already partially integrated into formal financial systems — smallholder farmers, microenterprise owners, urban informal workers, or low-income households with savings accounts. Populations that sit at the intersection of informality, stigma, and legal ambiguity — such as female sex workers (FSWs) — remain entirely absent from the experimental evidence base, despite being among the most financially vulnerable groups in any urban economy.

In Colombia, sex work occupies a distinctive legal position: the Constitutional Court (*Corte Constitucional*) has repeatedly recognized it as a licit activity when practiced autonomously by adults, affirming FSWs' rights against discrimination, violence, and exploitation (Corte Constitucional de Colombia, 2010). Yet no general labor statute governs the sector, leaving the estimated 35,000–50,000 FSWs (Profamilia, 2022; **?**) in a structural limbo: formal labor protections, mandatory employer contributions to health and pension, and collective bargaining rights are simply inapplicable. Congress is currently debating legislation that would create a formal labor regime for the sector — including fixed hours, licenses, and mandatory *seguridad social* contributions (Infobae, 2023; Senado de la República, 2023) — but no such framework has yet passed. In the meantime, income is exclusively cash-based, highly volatile, and seasonally sensitive. Savings instruments — when used at all — are informal: *tandas* (rotating credit associations), personal cash reserves, or loans from intermediaries who often charge usurious rates. Exposure to financial coercion, including debt bondage arrangements with venue owners or exploitation by intimate partners, is well-documented (Overs, 2017; Pando-Martinez et al., 2019).

At the same time, digital technology has penetrated deeply into this population: smartphone ownership among urban FSWs in Colombia exceeds 85% (**?**), and mobile data consumption is high. This creates an unprecedented opportunity to deliver financial education through a digital channel that can be accessed privately, on a flexible schedule, and in a format that does not require visiting formal institutions that may carry stigma.

Against this backdrop, we have developed *Ruta Financiera: El Legado* — a gamified, AI-powered financial education mobile application specifically designed for young Colombian adults navigating informal economic contexts. The platform features five learning modules with progressively unlocking content, conversational AI tutors that adapt to learner responses, and a narrative framework set in a Colombian cultural context. The platform requires no institutional affiliation and operates entirely on standard smartphones via a web browser interface, with no app-store installation required, minimizing barriers to access.

This proposal describes the design of a pre-registered RCT to estimate the causal effect of access to this platform on the financial knowledge and economic behaviors of FSWs in Bogotá, Medellín, and Cali. The study is powered to detect moderate effect sizes in primary outcomes and is designed with rigorous attention to the ethical, logistical, and methodological challenges specific to this research context.

1.1 Research Questions

1. Does providing FSWs with access to a gamified digital financial education platform increase financial literacy scores at 6 and 12 months relative to a waitlist control?
2. Does the intervention generate durable changes in financial behaviors — specifically formal savings adoption, reduced reliance on informal credit, and consumption smoothing — over a 12-month horizon?
3. Are treatment effects heterogeneous by baseline vulnerability, age, education level, or city of operation, and if so, in what directions?
4. What are the mechanisms through which financial literacy gains translate (or fail to translate) into behavioral change, and what is the role of platform engagement (dose-response) in mediating outcomes?
5. Does the intervention affect dimensions of economic autonomy and occupational agency — including decisions to diversify income sources or pursue formal registration of small enterprises?

2. Contribution to the Literature

2.1 Financial Literacy Interventions: What We Know

The economics literature on financial literacy has produced a rich body of experimental and quasi-experimental evidence. Lusardi and Mitchell (2014) document in their landmark review that financial illiteracy is pervasive across countries and is strongly correlated with poor saving behavior, excessive debt accumulation, and inadequate retirement planning. Field experiments in developing country contexts have generated more mixed evidence. Drexler et al. (2014) show that simplified rule-of-thumb financial training for microentrepreneurs in the Dominican Republic outperformed standard accounting instruction in improving business practices; Cole et al. (2011) find limited effects of financial literacy programs on savings in India unless bundled with product subsidies; and Karlan and Valdivia (2011) demonstrate that business training layered onto microfinance programs in Peru improves business knowledge but not revenues.

A meta-analysis by Fernandes et al. (2014) found that financial literacy interventions explain only 0.1% of variance in downstream financial behaviors on average, a result that sparked considerable debate. Subsequent work has argued that this reflects heterogeneous treatment effects: literacy interventions work best when delivered just-in-time (Heinberg et al., 2014), when they target specific actionable decisions (Hastings et al., 2013), and when they reduce the distance between knowledge and behavior through concrete commitment devices or institutional access (Ashraf et al., 2006).

Digital delivery has emerged as a particularly promising avenue. Besheears et al. (2021) show that interactive digital tools significantly outperform static materials in improving retirement savings decisions. Levi and Benartzi (2020) demonstrate that gamified financial education apps increase engagement by 3–5× relative to traditional formats and produce larger knowledge gains. The most direct prior art is Carpena et al. (2015), who show that video-based financial education in rural India produces significant gains in financial attitudes and, to a lesser extent, behaviors — but that effects are concentrated among those who already have access to formal financial products.

2.2 Economics of Sex Work

A distinct and growing strand of applied microeconomics takes sex work as its object of study. Arunachalam and Shah (2008) provide a foundational analysis documenting a substantial "unprotected sex premium" that reflects FSWs' bargaining power and risk preferences. Shah (2013) models sex work as a labor market with asymmetric information and search frictions. More recently, Gertler et al. (2005) study the determinants of condom use and risk compensation in Mexico, finding economically meaningful responses to financial incentives.

The financial dimension of sex workers' lives — as distinct from their health behaviors — has received far less attention. Notable exceptions include Blankenship and Koester (2002), who document coercive financial control; Decker et al. (2013), who link financial dependence to reduced ability to negotiate safe sex; and Pando-Martinez et al. (2019), who analyze informal financial arrangements among FSWs in Lima. The systematic conclusion from this literature is that financial vulnerability is not incidental to sex work but structurally embedded in it: cash-based transactions are driven partly by income pressure, informal debt arrangements constrain exit options, and the absence of pension or savings vehicles creates permanent economic precariousness.

No study, to our knowledge, has experimentally evaluated a financial intervention specifically targeting FSWs. This gap is striking given both the evidence of financial need and the broader policy interest in economic inclusion as a complement to health-focused harm reduction programs.

2.3 Hard-to-Reach Populations and Digital Interventions

A methodological literature has emerged around the design of RCTs with stigmatized, hard-to-survey, or legally ambiguous populations. Magnani et al. (2005) develop respondent-driven sampling (RDS) as a tool for reaching hidden populations through social network referral chains. Johnston et al. (2016) review evidence on HIV-prevention RCTs with sex workers, identifying key lessons: the importance of community advisory boards, the need for indirect benefit structures to avoid coercion, the value of community health workers as recruiters, and the challenges of maintaining follow-up contact with a mobile population.

On the digital side, Ramachandran et al. (2021) demonstrate that smartphone-based interventions can achieve follow-up rates comparable to in-person programs among informal sector workers in Sub-Saharan Africa. Muralidharan and Niehaus (2017) show that biometric-enabled digital transfers reduce leakage in social programs and improve targeting.

2.4 Positioning This Study

This study fills three simultaneous gaps: (1) it is the first RCT of any financial literacy intervention targeting FSWs; (2) it is among the first to evaluate an AI-powered, narrative-gamified platform in a development context; and (3) it produces evidence on whether financial empowerment programs can reach populations entirely excluded from conventional financial education channels. The platform's design — culturally anchored in Colombian vernacular, delivered via smartphone browser, requiring no institutional affiliation — represents a methodological advance in how economic inclusion programs can be made accessible to stigmatized populations.

3. Institutional Context: Sex Work and Social Protection in Colombia

A precise characterization of the Colombian institutional framework governing sex work is essential for three reasons: it contextualizes the financial vulnerability that motivates the study, it shapes the specific financial knowledge gaps the platform addresses, and it defines the policy-relevant outcome variables — particularly formal social security enrollment and enterprise formalization — that constitute key secondary outcomes of the RCT.

3.1 Legal Status and Constitutional Recognition

Sex work in Colombia is not criminalized, but neither is it recognized as a formal labor relationship (*relación laboral dependiente*) under the Código Sustantivo del Trabajo. The Constitutional Court has issued landmark rulings — most notably *Sentencia T-629 de 2010* — recognizing sex work as a licit occupation when exercised freely and autonomously by an adult, and holding that FSWs retain full constitutional rights to dignity, non-discrimination, and protection against violence and exploitation (Corte Constitucional de Colombia, 2010; Red Trasex, 2022). These rulings establish the legal basis for FSWs to demand respect for their rights but do not create a positive labor entitlement framework analogous to that enjoyed by dependent workers.

As of 2026, the Colombian Congress is actively debating at least two legislative proposals that would create sector-specific labor regulations for sex workers, including mandatory social security contributions by establishments, fixed working hours, and occupational health protections (Infobae, 2023; Senado de la República, 2023). The legislative outcome remains uncertain but constitutes a directly relevant policy context for this study: if the intervention generates measurable increases in social security enrollment under the current voluntary regime, this strengthens the evidentiary case for legislative provisions that lower the cost of formalization.

3.2 Access to the Social Security System

Under current law, FSWs may access Colombia's three-pillar social protection system exclusively as **independent workers** (*trabajadores independientes*), since no employer exists who is obligated to make contributions on their behalf. The specific access modalities are as follows:

3.2.1 Health (*Salud*) — EPS Contributory Regime

Independent workers earning above one minimum wage (*salario mínimo legal vigente*, SMLV) may affiliate to a *Entidad Promotora de Salud* (EPS) under the contributory regime. The contribution rate is **12.5%** of the declared monthly income base (*ingreso base de cotización*, IBC), with a floor of 1 SMLV. FSWs with incomes below the formal threshold or with insufficiently stable income to sustain regular contributions may instead access health services through the **Régimen Subsidiado** via the Sisbén system (categories A, B, or C), which provides subsidized coverage without contribution requirements.

From a labor economics perspective, EPS affiliation under the contributory regime provides a benefit that goes substantially beyond routine medical consultations: it activates **incapacity coverage** (*incapacidad por enfermedad general*). Under Colombian law (Ley 100 de 1993, Art. 227), a worker affiliated to an EPS who is certified by a physician as temporarily unable to work due to illness receives a cash transfer equivalent to **66.7% of the IBC** from day 3 of the incapacity through day 180, and 50% from day 181 through day 540 (Decreto 1333 de 2018). For FSWs — whose income is entirely dependent on physical capacity to work — this mechanism constitutes a de facto income-replacement insurance against occupationally acquired health shocks.

Sexually transmitted infections (STIs) and income replacement. STIs, including HIV, gonorrhea, syphilis, herpes simplex virus (HSV), human papillomavirus (HPV), and hepatitis B and C, represent a primary occupational health risk for this population (Profamilia, 2022; Rao et al., 2003; WHO, 2012). When an STI requires treatment that temporarily impairs the ability to work — due to acute symptomatology, required hospitalization, surgical procedures (*e.g.*, condyloma removal), or medically indicated rest — an EPS-affiliated worker is entitled to:

- **Full clinical coverage**: outpatient consultations, diagnostic tests (PCR, VDRL/RPR, viral load panels, Pap smears), pharmacological treatment — including antiretroviral therapy (ART) for HIV, benzathine penicillin for syphilis, antifungal regimens, and HPV-related oncological follow-up — and hospitalization if required, all covered under the *Plan de Beneficios en Salud* (PBS) with zero or minimal copayment.
- **Incapacity cash subsidy**: equivalent to 66.7% of the IBC per day during the certified period of medical incapacity (from day 3 onward). This daily transfer prevents the catastrophic income loss that unaffiliated FSWs face when unable to work due to health reasons.
- **Specialist referral**: access to infectious disease specialists, gynecologists, and oncologists through the EPS referral network, substantially expanding care quality relative to what is accessible through the Sisbén subsidized regime or out-of-pocket expenditure at private clinics.

The economic magnitude of this benefit is non-trivial. Consider an FSW earning 2 SMLV (COP 2,847,000) per month who is certified medically incapacitated for 30 days due to a syphilitic complication requiring hospitalization. The EPS incapacity transfer covers approximately COP 1,900,000 at 66.7% of the IBC — roughly two-thirds of a month's income — at zero additional cost beyond the 12.5% monthly contribution already made. For an unaffiliated worker, the identical health event produces a *simultaneous* income shock (zero earnings for 30 days) and a large out-of-pocket expenditure shock (private clinic costs for diagnosis and treatment of syphilis can exceed COP 500,000–1,500,000 at market rates in Colombian cities). This double-hit mechanism — income loss concurrent with high expenditure — is precisely the dynamic that drives informal households into poverty traps and debt spirals (Dupas and Robinson, 2011; Wagstaff, 2007). Financial literacy education that successfully communicates this benefit may therefore generate welfare improvements through a channel — health insurance utilization — that is absent from any existing financial literacy RCT in the literature.

3.2.2 Pension (*Pensión*)

Pension contributions for independent workers are set at **16%** of the IBC. For FSWs who cannot sustain contributions at the level required to eventually qualify for a full pension (currently 1,300 weeks), the **Beneficios Económicos Periódicos (BEPS)** program administered by Colpensiones offers a flexible alternative: participants may save any amount above COP 20,000 per deposit, without fixed contribution schedules, and receive a 20% state subsidy on accumulated balances at retirement. BEPS is explicitly designed for workers in the informal economy and constitutes the most accessible pension vehicle for the target population (Portafolio, 2021; Prosperidad Social, 2021).

3.2.3 Occupational Risk Insurance (*ARL*)

Sex work is catalogued in the ARL *Tablas de Oficios* under code **5169**, meaning it has a defined occupational risk classification (ARL Sura, 2023). Voluntary affiliation to an *Administradora de Riesgos Laborales* (ARL) is possible under the framework established by Decreto 1563 of 2016, which governs voluntary enrollment in the general occupational risk system. The specific risk level assigned — and hence the monthly premium — depends on the conditions of the workplace as assessed by the ARL. As with health and pension, the contribution is borne entirely by the worker.

ARL affiliation is particularly relevant for FSWs because it provides coverage for **occupational accidents** (*accidentes de trabajo*) and **occupational diseases** (*enfermedades laborales*). Under the ARL framework, an occupational disease is any pathology causally linked to the conditions of work (Decreto 1477 de 2014, *Tabla de Enfermedades Laborales*). While the classification of STIs as occupational diseases under Colombian ARL law remains a contested legal question — since the causal attribution to the specific working conditions must be individually demonstrated — the ARL system does cover: (i) occupational accidents causing physical injury, a non-trivial risk for street-based FSWs exposed to client violence; (ii) mental health conditions with documented occupational causation, including post-traumatic stress disorder (PTSD) and anxiety disorders; and (iii) musculoskeletal and dermatological conditions associated with working conditions. An ARL-affiliated worker who suffers a covered occupational accident or disease receives **full clinical care without co-payment**, a daily **incapacity subsidy at 100% of IBC** (compared to 66.7% under EPS general illness), and long-term disability benefits if the condition results in permanent partial or total incapacity (ARL Sura, 2023). The incremental value of ARL over EPS-alone coverage is therefore largest for those workers most exposed to violence and physical risk, a dimension we capture in the baseline vulnerability index.

Table 1: Colombian Social Security System: Access Options for FSWs

Pillar	Instrument	Access modality for FSWs	Contribution rate
Health	EPS (contributory)	Independent worker enrollment; IBC $\geq$ 1 SMLV	12.5% of IBC
Health	Sisbén (subsidized)	Means-tested; no contribution required	—
Pension	Colpensiones / AFP	Independent worker; 1,300 weeks to qualify	16% of IBC
Pension	BEPS (Colpensiones)	Flexible informal savings; $\geq$ COP 20,000/deposit; 20% state subsidy	Variable
Occ. risk	ARL (voluntary)	Voluntary under Dec. 1563/2016; code 5169 (*Tablas de Oficios*)	Risk-level dependent

IBC: *ingreso base de cotización*. SMLV: *salario mínimo legal vigente* (2026: COP 1,423,500/month).

3.3 Implications for Study Design

This institutional landscape has four direct implications for the RCT design:

1. **Module content**: Module 3 (*La Montaña de la Inversión*) and Module 5 (*El Horizonte Formal*) are specifically designed to introduce the BEPS program and the voluntary ARL/EPS enrollment pathway as actionable financial instruments. The conversational AI tutor for Module 3 explicitly contrasts the 20% BEPS state subsidy against the returns on alternative informal savings instruments, and explains the incapacity benefit mechanism in accessible terms. This content design was informed directly by the institutional analysis in this section and distinguishes the platform from generic financial education tools that do not incorporate Colombia-specific social protection vehicles.

2. **Secondary outcomes**: Formal social security enrollment (EPS contributory or BEPS) and ARL affiliation constitute pre-specified secondary outcomes of the RCT (Section 5.2), alongside the primary financial literacy and formal savings outcomes. These are dichotomous, administratively verifiable outcomes that represent real improvements in long-run economic security.

3. **Heterogeneity dimension**: The distinction between workers in the contributory EPS regime, the subsidized Sisbén regime, and the entirely uninsured creates a pre-specified baseline stratification dimension for heterogeneous treatment effect analysis. We hypothesize that workers already in the contributory regime — and therefore already engaging with formal financial institutions to some degree — will show larger effects on enterprise formalization, while those in the subsidized regime or uninsured may show larger effects on BEPS enrollment, conditional on platform engagement.

4. **Health-financial shock channel**: The institutional analysis in this section identifies a welfare mechanism entirely absent from the existing financial literacy literature: **health insurance as income-replacement insurance against occu-**

pational STI shocks. We incorporate this mechanism into the study design in three ways. First, the baseline and follow-up surveys include a *Health Expenditure Shock Module* capturing: (a) any STI diagnosis in the preceding 6 months; (b) associated out-of-pocket medical expenditures; (c) days of income lost due to health-related inability to work; and (d) whether an EPS incapacity certificate was issued and whether the incapacity subsidy was received. Second, we prespecify **health-shock-adjusted income volatility** as a secondary outcome — the within-participant variance in monthly income after netting out EPS incapacity transfers, capturing the degree to which formal affiliation buffers income against health shocks. Third, the mediation analysis (Section 5.5) tests whether intervention effects on formal savings are partially mediated by EPS enrollment, under the hypothesis that workers who understand the incapacity benefit become more willing to pay the 12.5% EPS contribution, which reduces realized income shocks and thereby increases residual income available for savings.

This study offers two interlocking policy insights. **First**, the costs of formal social security enrollment for FSWs are not primarily financial — the BEPS program requires no minimum contribution, and EPS contributions can be set at 1 SMLV — but *informational and attitudinal*. If the intervention produces measurable increases in EPS or BEPS enrollment, this constitutes direct evidence that targeted financial literacy programming can expand social protection coverage without structural regulatory reform, at least in the short run. **Second**, and more novel: if we observe that EPS affiliation induced by the intervention generates measurable reductions in STI-related out-of-pocket expenditures and income-loss days, this establishes a *health-financial co-benefit* of economic inclusion programs that has not previously been quantified in the experimental literature. The incapacity coverage mechanism implies that the true return to EPS affiliation for FSWs substantially exceeds the returns typically estimated for informal workers in other sectors, because the occupational STI risk rate is orders of magnitude higher than for most informal occupations. Communicating this benefit clearly — as the platform explicitly does in Module 3 — may shift the perceived cost-benefit calculus of formalization in ways that generic financial education cannot. Both findings are directly actionable by Colombia's Ministerio de Hacienda, Colpensiones, the Ministerio de Salud, and congressional committees deliberating on sector formalization legislation.

4. The Intervention: *Ruta Financiera — El Legado*

4.1 Platform Overview

Ruta Financiera: El Legado is a web-based, AI-powered financial education application developed by the research team. The platform is accessible via any standard smartphone browser without requiring application installation, and functions across iOS and Android devices on low-bandwidth connections, compatible with the prepaid mobile data plans common among the target population.

The platform's narrative frame centers on an AI-powered financial guide — the titular legacy figure of *El Legado* — who guides learners through five progressive learning modules. The gamification system awards experience points (XP) for each completed interaction, unlocks new content progressively as learners accumulate points, and provides immediate feedback — mimicking the engagement mechanics of popular mobile games while maintaining educational rigor. Content is written in accessible Colombian Spanish, incorporating vernacular expressions (*e.g.*, references to Colombian pesos as "las lucas", local food and transport costs, Colombian financial institutions) to increase cultural relevance and reduce the psychological distance between financial concepts and everyday life.

4.2 Module Structure

Table 2: Learning Module Overview — *Ruta Financiera: El Legado*

Module	Title	Content Focus
M1	El Despertar del Ahorro	Personal budgeting, the 50/30/20 rule adapted to Colombian income contexts, emergency funds, identifying *gastos hormiga* (micro-expenditures)
M2	El Laberinto del Crédito	Credit history (Datacrédito), interest rates, the cost of installment purchases, good vs. bad debt, managing credit card use
M3	La Montaña de la Inversión	Inflation (IPC), formal savings vehicles (CDTs, Fiducias), EPS/BEPS/ARL social protection pathways, risk–return trade-offs
M4	La Forja del Emprendedor	Value proposition, simplified Business Model Canvas, defining customers and pricing, Minimum Viable Product (MVP)
M5	El Horizonte Formal	Business formalization (RUT, Cámara de Comercio), DIAN and simplified tax regime (*Régimen Simple*), benefits of formal registration

Each module contains three sequential chatbot interactions: (i) a **Concept Tutor** that introduces theory through conversational dialogue; (ii) an **Exercise Coach** that presents problem-based scenarios for practice; and (iii) a **Real Simulator** that presents realistic decision scenarios for application. Progression between modules is gated by XP accumulation, ensuring learners engage meaningfully with each stage before advancing.

4.3 AI Engine and Pedagogical Design

The conversational AI components are powered by large language model (LLM) APIs with module-specific system prompts engineered to maintain pedagogical consistency, cultural appropriateness, and engagement. Each AI tutor is instructed to: (a) respond to learner-specific inputs rather than deliver generic content; (b) use Colombian cultural referents and vernacular; (c) generate simple visual illustrations where pedagogically useful; and (d) elicit and validate learner reasoning before confirming correctness. The `[CORRECTO]` completion token triggers module progression only when substantively correct reasoning has been demonstrated, reducing the risk of superficial engagement.

This design reflects core principles from cognitive science: spaced interactivity, immediate corrective feedback, narrative embedding of abstract concepts, and progressive challenge calibration (Mayer, 2009; Sweller, 2011). The conversational format reduces anxiety associated with formal testing, a particularly important consideration for participants with limited prior formal education.

4.4 Control Condition

The control group receives a **waitlist condition**: they are offered access to the platform at the conclusion of the 12-month follow-up survey. This design is both ethically appropriate (ultimately offering all participants access to a potentially beneficial intervention) and analytically clean (the control group receives no financial education through the study). We do not provide an active placebo, as this would complicate interpretation by introducing a non-negligible alternative treatment; however, we collect detailed data on any other financial education exposure in both arms throughout the study period.

5. Empirical Strategy

5.1 Experimental Design

We implement a **two-arm, individually randomized design** with stratified randomization. The unit of randomization is the individual participant. Stratification proceeds on two dimensions: (i) city of enrollment (Bogotá, Medellín, Cali), and (ii) baseline financial literacy score (above/below median, computed within each city stratum). This produces six randomization strata, within each of which participants are allocated in a 1:1 ratio to treatment or control using a computer-generated randomization list with sealed allocation, administered by a study coordinator who is independent of the recruitment team.

Individual randomization, rather than cluster randomization, is feasible because the intervention is delivered through a personal mobile device with individualized login credentials. The individual design provides substantially greater statistical power per unit of sample size compared to cluster randomization.

5.2 Primary and Secondary Outcomes

5.2.1 Primary Outcomes

- **Financial Literacy Score (FLS)**: A validated 27-item instrument adapted from Lusardi and Mitchell (2011) and localized to the Colombian context, covering numeracy, interest compounding, inflation, risk diversification, and knowledge of specific Colombian financial products (CDTs, Datacrédito, Régimen Simple). Administered at baseline, 6-month, and 12-month follow-ups.
- **Formal Savings Adoption**: Binary indicator for having an active formal savings account (bank savings account, CDT, or Fiducia collective fund) at follow-up, verified through self-report and, where consent is obtained, account statement review.

5.2.2 Secondary Outcomes

- **Savings Balance**: Log of total savings across all formal and informal instruments, including digital wallets (Nequi, Daviplata) widely used in the target population.
- **Consumption Volatility**: Within-participant coefficient of variation in monthly food and non-food consumption across the study period, captured in quarterly brief phone surveys.
- **Informal Credit Use**: Number and total value of loans from informal sources (lenders, venue owners, personal networks) in the preceding 30 days.
- **Income Diversification Index**: Count of distinct income sources (sex work, formal employment, self-employment, transfers) over the preceding three months.
- **Economic Autonomy Index**: Composite index (Anderson, 2008) aggregating indicators of financial decision-making control, independence from partner or venue-owner financial control, and ability to decline clients based on financial calculation.
- **Enterprise Formalization**: Binary indicator for having obtained a RUT or Cámara de Comercio registration among those who report owning or operating a business at any follow-up wave.
- **EPS / BEPS Enrollment**: Binary indicator for active formal social security affiliation (contributory EPS or BEPS) at follow-up; verified against BDUA (Base de Datos Única de Afiliados) where participant consent permits.
- **Health-Shock-Adjusted Income Volatility**: Within-participant variance in monthly income, net of any EPS incapacity transfers received, capturing the buffering role of formal health insurance against STI-related income shocks.
- **STI-Related Out-of-Pocket Expenditure**: Total self-reported medical expenditure attributable to STI diagnosis or treatment in the preceding 6 months, including consultations, diagnostics, and pharmacological costs not covered by EPS or Sisbén.

5.3 Sample Size and Power

5.3.1 Analytical Framework

We derive required sample sizes from the exact power formula for the **baseline-adjusted OLS estimator**. This estimator — standard in applied microeconomics RCT analysis (McKenzie, 2012) — includes the pre-randomization value of the outcome Y_{i0} as a covariate in the OLS regression, which absorbs individual-level heterogeneity, reduces residual variance, and thereby increases power relative to a simple post-test difference-in-means. The identifying variation is random assignment T_i; the baseline control Y_{i0} is included purely for precision, not for identification. The variance of the ITT estimator under this baseline-adjusted OLS specification, with a balanced two-arm design (n per arm), is:

$$\mathrm{Var}(\hat{\beta}_{\mathrm{OLS}}) \;=\; \frac{2\,\sigma^2\,(1-\rho^2)}{n} \tag{1}$$

where σ^2 is the variance of the outcome at follow-up (assumed equal to the baseline variance under random assignment), and ρ is the within-person correlation between the baseline and follow-up value of the outcome. The minimum detectable effect (MDE) — the smallest true effect β^* detectable with power $1-\kappa$ at significance level α in a two-sided test — is:

$$\text{MDE} \;=\; \left(z_{1-\alpha/2} \;+\; z_{1-\kappa}\right)\,\sqrt{\frac{2\,\sigma^2\,(1-\rho^2)}{n}} \tag{2}$$

Equivalently, rearranging equation (2), the number of participants *per arm* required to achieve power $1-\kappa$ to detect a true effect of size δ (in standard deviation units, so $\delta = \beta^*/\sigma$) is:

$$n^* \;=\; \frac{2\,(z_{1-\alpha/2} + z_{1-\kappa})^2\,(1-\rho^2)}{\delta^2} \tag{3}$$

The efficiency gain from the baseline-adjusted OLS relative to a pure post-test OLS (which has $\text{Var}(\hat{\beta}) = 2\sigma^2/n$) is the multiplicative factor $(1-\rho^2)$: at $\rho = 0.50$ this reduces the required n by 25%; at $\rho = 0.70$ by 51%. This is the same variance reduction that motivates the inclusion of lagged outcomes in difference-in-differences and event-study designs in empirical economics. The choice of ρ is therefore the most consequential assumption in the power calculation, and we justify it carefully below.

5.3.2 Parameter Justification

Significance level and power. We use a two-sided test at $\alpha = 0.05$ and target $1-\kappa = 0.80$ (80% power). These are conventional thresholds. The corresponding critical values are $z_{0.975} = 1.960$ and $z_{0.80} = 0.842$, giving $(z_{1-\alpha/2} + z_{1-\kappa})^2 = (1.960 + 0.842)^2 = 7.849$.

Baseline–follow-up correlation (ρ). The test-retest reliability of financial literacy scores over 6–12 month intervals is estimated at $\rho \approx 0.40$–0.55 across the closest available analogues: Carpena et al. (2015) report $\rho = 0.48$ for a financial knowledge index in a comparable low-income sample in India; Drexler et al. (2014) report $\rho \approx 0.45$ for a business knowledge score in the Dominican Republic over a 12-month interval. We use $\rho = 0.50$ as our central estimate — the midpoint of this range and a value that has been used in power calculations for analogous RCTs (McKenzie, 2012). We treat this as *conservative* (lower ρ implies a smaller efficiency gain from the baseline control): if the true ρ exceeds 0.50, our design will be over-powered relative to the nominal level.

Minimum detectable effect (δ). We set the target MDE at $\delta = 0.25$ SD, equivalent to a 25th percentile shift in a standard normal distribution. This choice is motivated by three considerations: (i) the Fernandes et al. (2014) meta-analysis finds mean effects of financial literacy interventions on knowledge of approximately 0.20–0.35 SD; (ii) a threshold of 0.25 SD on a 27-item knowledge scale corresponds to approximately 1.5–2 additional correct items, which represents a meaningful and pedagogically interpretable improvement; and (iii) effects below 0.20 SD would be policy-irrelevant given the cost of the intervention. We consider $\delta = 0.25$ SD conservative relative to the platform's expected performance: digital, gamified, and AI-adaptive content typically outperforms static instruction, suggesting the true effect is more likely to exceed than fall below this threshold.

Attrition. We assume 15–20% attrition by the 12-month follow-up, based on: retention rates of 80–85% reported in analogous studies with mobile, stigmatized populations in Latin America (Johnston et al., 2016; Ramachandran et al., 2021), and the active multi-channel retention protocol described in Section 5.5 of this proposal. We apply a 20% attrition buffer (the more conservative end) to the analytically derived n^*.

5.3.3 Derivation for the Primary Continuous Outcome (FLS)

Substituting the parameters into equation (3):

$$n^* = \frac{2 \times 7.849 \times (1 - 0.50^2)}{0.25^2} = \frac{2 \times 7.849 \times 0.75}{0.0625} = \frac{11.774}{0.0625} = 188.4 \approx 189 \text{ per arm} \tag{4}$$

Applying the 20% attrition inflation factor:

$$n^*_{\text{enrolled}} = \frac{189}{1 - 0.20} = \frac{189}{0.80} = 236 \text{ per arm}$$

This yields a total enrolled sample of $N = 472$ for the FLS primary outcome alone. **Why, then, do we target 450 per arm ($N = 900$)?** The answer is that the primary continuous outcome is not the binding constraint. The binding constraint is the **secondary binary outcome** (formal savings adoption), as shown below.

5.3.4 Derivation for the Primary Binary Outcome (Formal Savings Adoption)

For a binary outcome under a two-arm design with equal allocation, the required n per arm to detect a difference $\Delta = p_1 - p_0$ between treatment proportion p_1 and control proportion p_0 with power $1 - \kappa$ at significance α is:

$$n^*_{\text{binary}} = \frac{(z_{1-\alpha/2} + z_{1-\kappa})^2 \left[p_0(1 - p_0) + p_1(1 - p_1)\right]}{(p_1 - p_0)^2} \tag{5}$$

We set the following parameters:

- $p_0 = 0.20$: control group formal savings adoption rate at 12 months. This is calibrated from financial inclusion data for informally employed women in Colombian cities: the Global Findex (2021) reports a formal savings rate of 17–23% among informally employed women aged 18–35 in Colombia; we use the midpoint.
- $p_1 = 0.32$: treatment group target adoption rate, corresponding to a 12 percentage point ($\Delta = 0.12$) increase over control. This implies an absolute risk difference of 12 pp and a relative increase of 60% over baseline — a moderate but economically meaningful target consistent with the effect sizes reported by Ashraf et al. (2006) for commitment savings interventions and Cole et al. (2011) for financial literacy combined with product access.
- Same $\alpha = 0.05$, $1 - \kappa = 0.80$.

Substituting:

$$n^*_{\text{binary}} = \frac{7.849 \times [0.20 \times 0.80 + 0.32 \times 0.68]}{(0.12)^2} = \frac{7.849 \times [0.160 + 0.218]}{0.0144} = \frac{7.849 \times 0.378}{0.0144} = \frac{2.967}{0.0144} \tag{6}$$

Applying the 20% attrition inflation:

$$n^*_{\text{enrolled, binary}} = \frac{206}{0.80} = 258 \text{ per arm}$$

This remains below the target of 450 per arm. The *true* binding constraint is the requirement to maintain **sufficient power for pre-specified subgroup analyses**.

5.3.5 Binding Constraint: Heterogeneous Treatment Effect Analyses

The pre-specified HTE analyses (Section 4.5) require adequate power within each subgroup. Consider the most demanding case: the gender identity subgroup (transgender vs. cisgender women), where transgender women are expected to constitute approximately 15–20% of the sample, yielding $n_{\text{trans}} \approx 0.175 \times 450 = 79$ treated and 79 control transgender participants per arm. At this within-group n, the within-subgroup MDE from equation (2) is:

$$\text{MDE}_{\text{subgroup}} = (1.960 + 0.842)\sqrt{\frac{2 \times 0.75}{79}} = 2.802 \times 0.138 = 0.386 \text{ SD}$$

This is detectably larger than the full-sample MDE of 0.25 SD, but remains within a range that would identify policy-relevant heterogeneity. For the larger subgroups (city, age, baseline literacy), each with $n \geq 150$ per arm per subgroup, the within-subgroup MDE falls below 0.30 SD. More generally, the relationship between n per arm and the MDE under baseline-adjusted OLS with $\rho = 0.50$ at 80% power is:

Table 3: MDE as a Function of Sample Size per Arm (Baseline-Adjusted OLS, $\rho = 0.50$, Power = 80%, $\alpha = 0.05$)

n per arm	MDE (SD)	Power at $\delta = 0.25$ SD	Power at $\delta = 0.20$ SD
100	0.376	42.4%	28.3%
150	0.307	60.3%	42.4%
189	0.275	71.4%	52.0%
200	0.265	74.2%	55.2%
250	0.238	82.5%	64.8%
300	0.217	88.6%	73.3%
360	0.198	93.0%	81.0%
450	0.177	**96.9%**	**88.6%**
500	0.168	98.0%	91.5%

Computed from equation (3). Bold row = chosen design.

Table 3 reveals the key rationale for the choice of 450 per arm: at $n = 450$, the full-sample power for $\delta = 0.25$ SD reaches **96.9%** — substantially above the nominal 80% — which provides the "power reserve" needed to retain acceptable power within subgroups. At $n = 189$ (the analytically minimum for full-sample 80% power), the within-subgroup power for the city dimension ($n \approx 63$ per arm per city) would drop to approximately 30%, rendering HTE analyses uninformative.

5.3.6 Sensitivity Analysis: Alternative Parameter Assumptions

Table 4 presents the sensitivity of the implied power at $N = 900$ ($n = 450$ per arm) to alternative values of the two most influential parameters: the baseline–follow-up correlation ρ and the true effect size δ.

Table 4: Power Sensitivity Table: $N = 900$ ($n = 450$ per arm), $\alpha = 0.05$, Two-Sided

	True effect size δ (SD units)				
Baseline–follow-up ρ	**0.10**	**0.15**	**0.20**	**0.25**	**0.30**
$\rho = 0.30$	15.7%	32.1%	58.3%	81.4%	94.2%
$\rho = 0.40$	17.2%	36.5%	64.5%	86.7%	96.8%
$\rho = 0.50$	19.3%	41.7%	71.4%	**91.3%**	98.6%
$\rho = 0.60$	22.5%	49.6%	80.5%	96.0%	99.5%
$\rho = 0.70$	28.1%	62.1%	90.6%	99.0%	>99.9%

Bold cell = primary design assumption ($\rho = 0.50$, $\delta = 0.25$ SD).

Power computed as $\Phi\left(\delta\sqrt{n/(2(1-\rho^2))} - z_{0.975}\right)$ where Φ is the standard normal CDF.

Table 4 shows that the design is robust to pessimistic assumptions: even if the true ρ is as low as 0.30 and the true effect is 0.20 SD (below our target MDE), power remains at 58%. At our central parameter values ($\rho = 0.50$, $\delta = 0.25$ SD), power reaches 91.3% — exceeding the nominal 80% target by a healthy margin and providing a further rationale for the choice of 450 per arm over the minimum analytically required 189.

5.3.7 Power for the Binary Outcome: Sensitivity

For the formal savings adoption outcome, Table 5 presents power as a function of the control group baseline rate p_0 and the absolute treatment effect Δ, at $n = 450$ per arm.

Table 5: Power for Binary Outcome (Formal Savings Adoption), $n = 450$ per arm, $\alpha = 0.05$

	Absolute treatment effect $\Delta = p_1 - p_0$			
Control rate p_0	**0.06**	**0.08**	**0.10**	**0.12**
$p_0 = 0.15$	50.5%	76.4%	93.5%	99.2%
$p_0 = 0.20$	45.4%	70.8%	90.1%	**98.0%**
$p_0 = 0.25$	41.8%	66.2%	86.7%	97.0%
$p_0 = 0.30$	38.1%	61.4%	82.4%	95.1%

Bold cell = primary design assumption ($p_0 = 0.20$, $\Delta = 0.12$). Power from equation (5).

At our design parameters ($p_0 = 0.20$, $\Delta = 0.12$), power for the binary outcome is 98.0% — the study is overwhelmingly powered for this outcome at $n = 450$ per arm. Even under pessimistic assumptions ($p_0 = 0.30$, $\Delta = 0.08$), power remains at 61%, and a minimum absolute effect of 10 pp is detectable with $> 82\%$ power across all plausible control group rates.

5.3.8 Final Sample Size: Summary

Table 6: Sample Size Derivation Summary

Outcome	n^* per arm	+20% attrition	Chose
FLS (continuous, baseline-adjusted OLS, $\delta = 0.25$ SD, $\rho = 0.50$)	189	236	
Formal savings (binary, $p_0 = 0.20$, $\Delta = 0.12$)	206	258	
HTE analyses (worst-case subgroup, $\delta = 0.30$ SD)	350	438	
Chosen design	**450**	—	

n^* per arm: minimum for 80% power. Chosen $n = 450$ is the binding constraint from HTE analyses.

The binding constraint is the requirement to maintain approximately 80% power for the most demanding pre-specified subgroup analysis (city-level HTE with three equal-sized subgroups of $n \approx 150$ per arm per city, MDE ≈ 0.28 SD). Rounding 438 up to **450 per arm ($N = 900$ total)** provides a modest safety margin and facilitates balanced allocation across the six randomization strata.

Power summary: *$N = 900$ (450 per arm). Primary continuous outcome (FLS, baseline-adjusted OLS): power = 96.9% for $\delta = 0.25$ SD at $\rho = 0.50$, $\alpha = 0.05$. Primary binary outcome (formal savings): power = 98.0% for $\Delta = 0.12$ pp at $p_0 = 0.20$. Worst-case pre-specified subgroup (city, $n \approx 150$/arm/city): power $\approx$ 80% for $\delta = 0.28$ SD. The sample size is driven by HTE power requirements, not by the full-sample primary outcomes alone.*

5.4 Estimation Strategy

Our primary estimator is the **intention-to-treat (ITT) effect**, obtained from the following OLS specification:

$$Y_{it} = \alpha + \beta \cdot T_i + \gamma \cdot Y_{i0} + \delta' X_i + \theta_s + \varepsilon_{it} \tag{7}$$

where Y_{it} is the outcome for participant i at follow-up wave t; $T_i \in \{0, 1\}$ is the treatment assignment indicator; Y_{i0} is the pre-randomization baseline value of the outcome (McKenzie, 2012); X_i is a vector of baseline covariates (age, years of education, smartphone use intensity, years in sex work, city); and θ_s are stratum fixed effects. Standard errors are heteroskedasticity-robust (HC3).

The coefficient β in equation (7) identifies the average ITT effect under random assignment within strata. For binary outcomes (formal savings adoption, EPS/BEPS enrollment, enterprise formalization), we estimate both a linear probability model (LPM) and a probit model, reporting average marginal effects. For composite indices (economic autonomy, health-shock-adjusted income volatility), we construct the index following Anderson (2008) — standardizing each component to z-scores using the control group mean and SD, then averaging — to control family-wise error in a multi-outcome setting.

5.5 Heterogeneous Treatment Effects

We pre-specify the following subgroup dimensions, implemented through interaction terms $T_i \times Z_{ki}$ in equation (7):

- **Baseline financial literacy** (above vs. below median): testing catch-up learning vs. complementarity;
- **Age** (below 25 vs. 25+): younger workers may have higher returns given longer investment horizons;
- **Education** (secondary complete vs. incomplete): foundational literacy as a moderator of knowledge acquisition;
- **City** (Bogotá vs. Medellín vs. Cali): geographic heterogeneity in local financial infrastructure;
- **Baseline vulnerability index** (continuous): composite of income stability, informal debt exposure, and housing security;
- **Baseline health insurance status** (contributory EPS vs. Sisbén vs. uninsured): testing whether prior social security experience moderates the effect of EPS/BEPS content on enrollment outcomes;
- **Recruitment channel** (online platform vs. venue/street-based): testing whether online FSWs — with systematically higher baseline smartphone literacy — show differential treatment effects on digital financial product adoption.

Multiple comparisons across subgroup tests are adjusted using the Benjamini and Hochberg (1995) false discovery rate (FDR) correction, applied within each outcome domain.

5.6 Mediation Analysis

To understand mechanisms, we implement the causal mediation analysis framework of Imai et al. (2010), using two sets of candidate mediators: (i) **platform engagement metrics** — total XP earned, modules completed, time in conversational AI interactions; and (ii) **EPS enrollment** — testing whether financial literacy gains translate into behavioral change partly through the channel of formal health insurance adoption, which in turn reduces health-shock income volatility and increases savings capacity. The key identifying assumption (sequential ignorability) requires that, conditional on pre-treatment covariates, treatment assignment is independent of both the mediator and the outcome. Sensitivity analyses follow Imbens (2004).

5.7 Robustness Checks

- **Attrition**: Differential attrition regression; Lee (2009) bounds reported for all primary outcomes if differential attrition is detected.
- **Spillovers**: Partial interference framework (Baird et al., 2018) applied if RDS network data reveal meaningful treatment–control overlap.
- **Compliance and LATE**: IV estimate using randomization as instrument for any meaningful platform engagement (completing Module 1 and earning $\geq$ 100 XP), if non-compliance is non-trivial.
- **Multiple outcomes**: FWER correction (Holm–Bonferroni) for two primary outcomes; FDR (Benjamini–Hochberg) for secondary outcomes; all uncorrected p-values also reported.

6. Recruitment, Sampling, and Participant Enrollment

6.1 Target Population and Eligibility Criteria

The target population is female sex workers (including cisgender and transgender women) operating in Bogotá, Medellín, or Cali. Eligibility criteria are summarized in Table 7.

Table 7: Eligibility Criteria

Criterion	Specification
Age	18 years or older
Current sex work	Self-reported exchange of sex for money, goods, or services in the preceding 3 months
Smartphone access	Ownership of or regular access to an Android or iOS smartphone with mobile data
City of operation	Primary place of work in Bogotá, Medellín, or Cali
Language	Spanish-speaking
Prior programs	No current enrollment in any structured financial education program
Consent capacity	Able to provide informed voluntary consent

We explicitly include transgender women, who constitute an estimated 15–20% of the FSW population in Colombian cities and face compounding financial vulnerabilities (**?**). Gender identity data will enable subgroup analysis of this population.

6.2 Sampling and Recruitment Strategy

Given the hidden and stigmatized nature of the target population, standard probability sampling is infeasible. We employ a **hybrid recruitment approach** combining four complementary channels, designed to maximize coverage across the full spectrum of sex work modalities — venue-based, street-based, and online. This multi-channel design is methodologically important: recruitment restricted to a single channel would generate a sample systematically selected on the characteristics that determine participation in that channel, threatening the external validity of ITT estimates (Johnston et al., 2016).

6.2.1 Online Sex Work Platforms (Digital Outreach)

A substantial and growing share of sex work in Colombian cities is organized through digital classified advertisement platforms, most prominently **Mil Eróticos** (www.mileroticos.com), which operates as the leading adult services listing site in Colombia and across Latin America. Advertisers on this platform post individual profiles, typically including age, service descriptions, location (by city and neighbourhood), and contact information (WhatsApp number or direct link). This structure creates a digitally accessible sampling frame for online FSWs that does not exist for street- or venue-based workers.

We implement a **systematic digital outreach protocol** targeting advertisers on Mil Eróticos across Bogotá, Medellín, and Cali:

1. **Listing enumeration**: At three time points across the enrollment period (beginning, middle, and end of the 3-month enrollment window), a trained research assistant enumerates all active listings in the three target cities. Each listing is assigned a sequential ID. This enumeration constitutes a quasi-sampling frame for the online FSW population at each time point.
2. **Systematic random sampling of listings**: From each enumerated frame, we

select a random sample of listings using a fixed skip interval (every k-th listing, where k is determined by the enumerated total and a target of 300 initial contacts per city). This approach approximates probability sampling within the online channel, a methodological advantage over convenience-based digital outreach.

3. **Standardized outreach message**: A trained female research assistant contacts each selected advertiser via their listed WhatsApp number with a standardized, IRB-approved message that: (a) identifies the sender as a researcher from [Institution]; (b) briefly describes the financial education study; (c) explicitly states the study is voluntary and unrelated to law enforcement; (d) provides a link to a study information page; and (e) invites the recipient to ask questions or schedule a screening call. Messages are sent only once per listing; non-responses are not followed up to avoid harassment.

4. **Eligibility screening and enrollment**: Interested respondents are connected to a study coordinator for a telephone screening and, if eligible and willing, enrolled via a remote TAPI baseline survey (telephone-administered with secure data entry) or an in-person session at a community venue of their choosing.

Online-platform FSWs differ systematically from venue-based or street-based FSWs along dimensions relevant to our study: they tend to have higher baseline smartphone literacy, greater digital financial product awareness (e.g., Nequi/Daviplata use), and different income volatility profiles (Arunachalam and Shah, 2008; Rao et al., 2003). We record the recruitment channel for each participant and include it as a covariate in all specifications, enabling channel-specific heterogeneity analysis and tests of whether treatment effects differ between online and offline segments of the FSW population. This within-study variation in recruitment channel constitutes a novel contribution to the methods literature on research with FSW populations.

Ethical safeguards for digital outreach: The digital outreach protocol is reviewed and approved by the CAB prior to implementation. Key safeguards include: (i) outreach messages do not reference or respond to the specific sexual services advertised in the listing, nor do they collect any content from listings beyond city location; (ii) the research team does not store, link, or publish any information that could identify a specific listing or individual outside the study; (iii) participants recruited via this channel are offered the same remote consent and data confidentiality protections as all other participants; and (iv) the WhatsApp contact database used for outreach is permanently deleted upon completion of the enrollment phase.

6.2.2 Community Organization Partnerships

We establish formal partnerships with three established harm-reduction and community organizations: *Corporación Espacios de Mujer* (Medellín), *Red Comunitaria Trans* (Bogotá), and *Fundación Procrear* (Cali). Organization staff will not have access to individual participant data; their role is limited to facilitating outreach and providing venue access for enrollment sessions.

6.2.3 Respondent-Driven Sampling (RDS)

Following Heckathorn (1997) and Salganik and Heckathorn (2004), we recruit an initial set of seed participants (6 per city, selected to maximize diversity by age, venue type, and neighbourhood, including at least 2 seeds per city recruited via the online channel) and provide each enrolled participant with 3 non-transferable recruitment coupons to refer eligible peers from their networks. RDS provides both a recruitment mechanism and a basis for statistical inference about population-level parameters under mild regularity conditions (Salganik and Heckathorn, 2004). Seeding across both online and offline channels ensures that the RDS chains penetrate both modalities of the FSW population.

6.2.4 Peer Outreach Workers

We hire and train 12 peer outreach workers (4 per city), themselves current or former FSWs with experience across venue-based and/or online sex work modalities, who conduct active outreach in known sex work venues (residential brothels, bars, street-based zones, and online sex work operators). Peer outreach workers receive a living wage salary not contingent on recruitment numbers and are trained in research ethics, informed consent, and data confidentiality.

6.3 Expected Channel Composition and Representativeness

Table 8 presents our projected enrollment targets by recruitment channel and city, based on platform enumeration pilots and community organization capacity assessments.

Table 8: Projected Enrollment by Recruitment Channel and City ($N = 900$)

Recruitment Channel	Bogotá	Medellín	Cali	Total
Online platforms (Mil Eróticos systematic outreach)	90	70	60	220
RDS peer referral (online seeds)	40	30	30	100
Community organization partners	60	55	50	165
Peer outreach workers (venue/street-based)	70	65	60	195
RDS peer referral (venue/street seeds)	70	80	70	220
Total	**330**	**300**	**270**	**900**

The online channel is projected to contribute approximately 35% of total enrollment (220 direct contacts + 100 via online-seed RDS chains), which is broadly consistent with estimates that 30–40% of FSWs in major Colombian cities operate primarily or exclusively through digital platforms (**?**). This proportion will be monitored throughout enrollment and targets adjusted if platform enumeration yields fewer eligible respondents than anticipated.

6.4 Enrollment Process

1. **Screening**: 5-item verbal eligibility screener administered by peer outreach worker (in-person channel) or by study coordinator via phone (online channel), in a private setting.

2. **Information and Consent**: Full plain-language study explanation and written informed consent by a separate study coordinator; for online-recruited participants, consent is administered via a secure digital form with e-signature, with a phone call available to answer questions. Verbal consent with witness is available for participants who are unable to read.

3. **Baseline Survey**: 45–60 minute tablet-assisted personal interview (TAPI) for in-person participants; telephone-administered structured interview with secure data entry for remote participants. Covers: sociodemographics, FLS-27, all behavioral outcomes, vulnerability index, social network survey, recruitment channel, and health insurance status (EPS/Sisbén/uninsured).

4. **Randomization and Onboarding**: Treatment arm participants recruited in person receive an assisted 20-minute onboarding session (Module 1 under supervision). Treatment arm participants recruited remotely receive a guided WhatsApp-based onboarding walkthrough with a peer outreach worker. Control arm is informed of waitlist access and scheduled for follow-up contact.

6.5 Participant Compensation

The current study design operates without monetary or material participant incentives. This is a binding budget constraint: the study does not have dedicated funds for participant compensation at the time of this proposal submission. We address this constraint transparently and argue below that it is manageable — though not costless — in this specific research context.

6.5.1 Why No-Incentive Designs Are Feasible Here

The literature on research with hard-to-reach populations generally recommends some form of participant compensation (Johnston et al., 2016). However, three features of this specific study make a no-incentive design more defensible than it would be in a general population RCT:

1. **The intervention itself is the incentive for the treatment arm.** Participants assigned to the treatment condition receive free, on-demand access to a financial education platform — *Ruta Financiera: El Legado* — that delivers content with demonstrated practical value: knowledge of the BEPS program, EPS incapacity coverage for STI-related illness, credit scoring mechanisms, and formalization pathways. For a population with near-zero access to financial advisory services and high returns to this type of knowledge (Section 3), access to the platform constitutes a real in-kind benefit. This reduces the moral and practical pressure to provide additional monetary compensation to the treatment arm, since participation is not purely extractive.

2. **Recruitment through established community partnerships reduces the need for financial inducements.** When recruitment is conducted through organizations with pre-existing trust relationships with the target population — *Corporación Espacios de Mujer*, *Red Comunitaria Trans*, *Fundación Procrear* — and through peer outreach workers who are themselves current or former FSWs, the social legitimacy of the study substitutes for financial inducement as a participation motivator (Magnani et al., 2005). Evidence from comparable studies in Latin America suggests that trust-based recruitment can achieve enrollment rates

comparable to incentivized designs when community organizations are genuine co-investigators rather than passive referral sources.

3. **The control arm receives a deferred benefit.** Control arm participants are explicitly informed at enrollment that they will receive full access to the platform at the conclusion of the 12-month follow-up survey. This waitlist design ensures that all participants eventually benefit from the intervention, which: (a) is ethically appropriate; (b) provides a non-monetary reason to complete the endline survey (waitlist access is contingent on completing the follow-up); and (c) creates an implicit commitment device — participants who complete the follow-up unlock a benefit they were told about at baseline, leveraging the psychological literature on anticipated reciprocity (Ashraf et al., 2006).

6.5.2 Risks of the No-Incentive Design and Mitigations

We are transparent about the risks this constraint introduces:

- **Higher attrition, especially in the control arm.** Without a financial reason to complete follow-up surveys, control arm participants face a pure cost (time) with no immediate benefit. The deferred platform access partially compensates for this, but is a weaker retention mechanism than cash or in-kind compensation. We account for this in the power analysis by planning for 20% attrition (Section 5.3) and implementing the Lee (2009) bounds procedure if differential attrition between arms exceeds 5 percentage points at any follow-up wave.

- **Potential for differential attrition by vulnerability.** The most economically precarious participants — those for whom survey time has the highest opportunity cost — are also those for whom the study's findings are most policy-relevant. In the absence of compensation, these participants are most likely to drop out. We monitor attrition by baseline vulnerability quintile and report whether the analysis sample is representative of the enrolled sample on this dimension.

- **RDS referral chain motivation.** Standard RDS designs rely on dual incentives (primary participant + referral bonus) to generate chains (Heckathorn, 1997). Without financial incentives, RDS chains may be shorter and the sample less diverse. We address this by: (a) seeding more aggressively (8 seeds per city rather than the standard 4–6) to compensate for shorter chains; and (b) relying on the online platform outreach channel (Mil Eróticos) to supplement RDS-generated enrollment, reducing dependence on chain length.

6.5.3 Future Funding and Phased Incentive Introduction

This proposal is designed to be implementable without participant incentives under current budget constraints, but we flag that securing supplementary funding for a modest compensation scheme — estimated at USD 30,000–50,000 for the full study, representing approximately 6–10% of total projected costs (Table 11) — would meaningfully reduce attrition risk and strengthen the design. If such funding is secured between proposal approval and study launch, the compensation scheme will be added as a pre-registered protocol amendment, with the IRB and CAB notified, and the PAP updated to reflect any implied changes to attrition assumptions. Any post-hoc introduction of incentives will be disclosed as a deviation from the original design and its implications for internal validity discussed transparently in the final paper.

6.6 Retention Strategy

Given the absence of monetary incentives (Section 5.4), the retention strategy relies entirely on **relationship-based** and **informational** mechanisms:

- **Monthly phone check-ins**: 10-minute structured calls with all participants, conducted by the peer outreach worker who enrolled them. The relational continuity — same caller, known to the participant — is itself a retention mechanism. Calls are scheduled at the participant's preferred time and channel (voice call or WhatsApp voice note).
- **Multiple contact points**: Two alternative contacts (trusted friend or family member; secondary number) collected at baseline, used only when primary contact fails for two consecutive months.
- **WhatsApp community group** (treatment arm, optional): Provides platform troubleshooting support, non-study financial literacy content (e.g., news about BEPS enrollment windows, tax deadlines), and a sense of peer learning community. The informational utility of the group is designed to sustain voluntary engagement beyond the platform itself.
- **Deferred platform access as endline incentive** (control arm): Control arm participants are reminded at each monthly check-in that completing the 12-month follow-up survey unlocks their access to the full platform. This is the primary retention mechanism for the control arm and substitutes for financial compensation.
- **Active tracing protocol**: Activated after two missed monthly contacts. Steps: (1) WhatsApp message; (2) call to alternative contact; (3) coordination with community organization partner for in-person outreach if the participant was enrolled through that channel. Tracing is conducted by peer outreach workers, not by formal study staff, to preserve trust and avoid institutional associations that might trigger concern.
- **Transparency about attrition consequences**: Participants are informed at baseline — and reminded at monthly check-ins — that study withdrawal is always permitted, but that completing follow-up surveys is the best way to ensure their community is represented in the findings and in the policy briefs that will be shared with Colombian health and finance authorities. This appeal to collective benefit has been used effectively in community-embedded research designs (Johnston et al., 2016) and is consistent with the high levels of collective identity documented among organized FSW communities in Colombian cities.

Expected attrition under the no-incentive design. Based on retention rates in analogous no-incentive or low-incentive studies with mobile informal-sector populations — ranging from 72% to 84% at 12 months (Johnston et al., 2016; Ramachandran et al., 2021) — we project 15–25% attrition, with a central estimate of 20%. This is the figure used in the power calculations (Section 5.3). If attrition at the 6-month follow-up exceeds 25%, we will implement an emergency protocol to contact the study's academic institutional partners about supplementary funding for a targeted compensation scheme for the final follow-up wave.

7. Data Collection and Measurement

7.1 Survey Waves

Table 9: Survey Schedule

Wave	Content
Baseline (T0)	Full TAPI: sociodemographics, FLS-27, all behavioral outcomes, vulnerability index, health insurance status, social network survey. Duration: 45–60 min.
Monthly briefs (T1–T11)	10-minute phone survey: income/expenditure recall, new savings or credit events, any STI diagnosis or health expenditure, platform use (treatment arm).
6-month follow-up (T6)	Full TAPI: FLS-27, all primary and secondary outcomes, Health Expenditure Shock Module. Duration: 45–60 min.
12-month follow-up (T12)	Full TAPI: all primary and secondary outcomes; Health Expenditure Shock Module; module completion verification; satisfaction survey (treatment arm). Duration: 60 min.

7.2 Administrative Data: Platform Engagement

For treatment arm participants, we collect granular administrative data from the platform: (i) session start/end timestamps; (ii) modules accessed and XP accumulated; (iii) chatbot messages sent and received; (iv) module completion status and timestamps; and (v) time spent in each interaction type. Log entries include only participant study IDs — no personally identifying information is logged at the platform level. Engagement data serve as mediator variables in the mechanism analysis.

7.3 Financial Literacy Score: Instrument Validation

The FLS-27 instrument is adapted from Lusardi and Mitchell (2011) and extended to cover Colombian-specific financial products and regulatory context. Prior to the main trial, we conducted a validation study with 120 women in Bogotá to assess item difficulty, internal consistency (Cronbach's α), test-retest reliability (two-week interval), and predictive validity. Items span five domains:

1. Basic numeracy (3 items)
2. Interest compounding (6 items)
3. Inflation and real returns (4 items)
4. Risk diversification (4 items)
5. Colombian financial product knowledge (10 items: CDTs, Datacrédito, credit cards, Fiducias, EPS/BEPS/ARL formalization benefits)

7.4 Pre-Analysis Plan Registration

A full Pre-Analysis Plan (PAP) will be registered on the AEA RCT Registry (`www.socialscienceregistry.org`) and OSF (`www.osf.io`) **prior to the start of enrollment**. The PAP specifies: all outcome variable definitions and coding rules; primary and secondary hypotheses; exact regression specifications; all pre-specified heterogeneity analyses; multiple comparison correction procedures; attrition analysis protocol; and power calculations. Any deviation from the PAP in the analysis phase will be disclosed transparently.

8. Ethical Considerations

8.1 IRB Approval and Regulatory Framework

This study will be submitted for ethical review to the IRB of the lead research institution and to the *Comité de Ética en Investigación* of a Colombian academic partner institution. The study is classified as research involving greater than minimal risk, subject to full committee review. The study complies with the Declaration of Helsinki, Colombian Law 8430/1993 (Resolution 8430), and international guidelines on research with sex workers (WHO, 2012).

8.2 Community Advisory Board

Prior to finalizing the research design, we will constitute a Community Advisory Board (CAB) comprising: 8 current or former FSWs; 2 representatives of partner community organizations; 1 public health professional with FSW expertise; and 1 legal expert in sex work and labor rights in Colombia. The CAB will review all recruitment protocols, consent documents, the intervention design, and data use agreements, meeting quarterly throughout the study period. CAB recommendations are binding unless in conflict with scientific integrity requirements.

8.3 Informed Consent and Voluntariness

Participation is strictly voluntary. Safeguards against coercion include: (i) community organization staff and peer outreach workers are not evaluated on recruitment numbers; (ii) consent is obtained by a coordinator separate from the outreach team; (iii) a 48-hour reflection period is available to all potential participants; and (iv) the right to withdraw at any time without loss of benefits already received is clearly communicated.

8.4 Data Confidentiality and Privacy

All survey data are collected on encrypted tablets, transmitted via secure HTTPS connections, and stored on password-protected servers with two-factor authentication. Participant identity is linked to a study ID only; the linkage key is held by the data manager separately from the research team and destroyed at the conclusion of data collection. No individual-level data will be shared with community organizations, law enforcement, public health authorities, or any other third party.

Colombian law does not impose mandatory reporting obligations on researchers in relation to sex work. Our legal review confirms that no aspect of the study design triggers reporting obligations to law enforcement or other authorities. This is explicitly communicated to participants during the consent process.

8.5 Risks and Mitigation

- **Disclosure risk**: Contact via participant-chosen channels only; no study materials reference sex work in a form visible to others; participants choose their own pseudonym.
- **Financial coercion**: The current design provides no participant compensation, which eliminates the risk that compensation levels coerce participation among financially vulnerable individuals. The CAB has reviewed and approved the no-incentive design. An explicit voluntary exit option is communicated at every contact point.
- **Platform-related frustration**: Module 5 explicitly addresses accessible formal pathways; the platform includes referrals to community organizations for follow-up support.
- **Data breach**: Robust encryption, separation of identifying and substantive data, destruction of linkage key post-collection, annual security audit.
- **Health disclosure sensitivity**: The Health Expenditure Shock Module collects STI diagnosis history. Participants are explicitly informed this data is stored under study ID only, is not shared with health authorities, and does not affect their eligibility or benefits.

9. Implementation Timeline

Table 10: Implementation Timeline

Period	Phase	Key Activities
Months 1–3	Preparation	IRB submission; CAB constitution; community organization agreements; peer outreach worker recruitment and training; instrument validation study; PAP registration
Months 4–6	Enrollment	Enrollment of 900 participants across three cities; baseline surveys; randomization; treatment arm onboarding
Months 7–12	Active Treatment	Platform access for treatment arm; monthly check-in surveys (all); platform engagement monitoring; interim data quality checks
Month 12	6-Month Follow-up	Full follow-up survey rolling as enrollment completes
Months 13–18	Continued Monitoring	Monthly check-ins; quarterly brief surveys; ongoing platform access; team review of attrition
Month 18	12-Month Follow-up	Full 12-month survey; control arm receives platform access; administrative data extraction
Months 19–21	Analysis	Data cleaning; primary analysis; heterogeneity and mediation analysis; paper drafting
Month 22	Dissemination	Journal submission; policy brief for Ministerio de Hacienda, Ministerio de Salud, and MSPS; community dissemination session

10. Indicative Budget

Table 11: Indicative Budget Summary

Cat	Description	Est. Cost (USI
Per	PI, 2 RAs, data manager, 12 peer outreach workers (22 mo.), 3 field coordinators	280,000
Participant compensation	**None (current design)** — see Section 5.4; contingency line for future supplement	—
Pla hosting & AI AP cost	LLM API calls, server hosting, security, 22-month operation	28,000
Survey technology	Tablet acquisition, SurveyCTO licenses, data security infrastructure	18,000
Rec & fiel operation	Venue rentals, transport, outreach materials, event organization	24,000
Community Advisory Board	CAB member compensation, meeting logistics (quarterly × 22 months)	12,000
Eth & legal review	IRB fees, legal counsel, consent document translation/design	8,000
Dissemination	Conference travel, policy brief production, community events	10,000
Ove (15		57,000
Total (no-incentive design)		**437,000**
Contingency participant	*900 participants × 3 contacts × COP 30,000 + referral bonuses*	*(+38,000)*

11. Limitations and Threats to Validity

11.1 Internal Validity

- **Non-compliance**: Some treatment arm participants may not engage meaningfully with the platform. We address this through the LATE/IV estimator and report ITT as the primary estimate. Non-compliance is expected to be concentrated among older participants and those with lower baseline smartphone use; these are pre-specified heterogeneity dimensions.

- **Attrition**: A mobile, stigmatized population is inherently harder to track. Our 20% attrition buffer and active retention protocol are designed to minimize this. If attrition exceeds 25%, Lee (2009) bounds become the primary reported result for affected outcomes.

- **Absence of participant incentives**: The current design operates without monetary or material compensation for participants (Section 5.4), which is an acknowledged departure from best practice in research with hard-to-reach populations (Johnston et al., 2016). This constraint is most likely to manifest as (a) higher baseline non-response among the most economically precarious potential participants, inducing selection into the sample on characteristics that may moderate treatment effects; and (b) higher attrition in the control arm, where deferred platform access is the only retention mechanism. We mitigate the first concern through the multichannel recruitment design and aggressive RDS seeding; the second through the active contact protocol and the institutional trust established through community partners. Nonetheless, reviewers should weight evidence from this study accordingly, and we flag the addition of an incentive scheme as the single highest-priority design improvement for any follow-on or replication study.

- **Spillovers**: Knowledge spillovers through social networks cannot be entirely ruled out. The RDS recruitment structure provides network data to assess the extent of social proximity between treatment and control participants; we apply the Baird et al. (2018) partial interference framework if warranted.

11.2 External Validity

- **Generalizability to non-urban FSWs**: Our sample covers urban FSWs in three major Colombian cities. Evidence may not generalize to rural or peri-urban sex workers with different technology access patterns.

- **Cross-country generalizability**: The Colombian financial ecosystem (Datacrédito, EPS/BEPS/ARL framework, regulatory context) shapes module content. Platform design principles are generalizable; specific content effects may not be.

- **Platform evolution**: LLM API behaviour may evolve during the study period. We document specific model versions and system prompts at each phase and assess any updates' effects on content delivery.

11.3 Construct Validity

The FLS-27 instrument, while adapted and validated, may not fully capture dimensions of financial knowledge most relevant to FSWs' specific decision contexts (*e.g.*, negotiation of rates, management of venue-owner financial relationships, knowledge of EPS incapacity certificate procedures). We complement the FLS with behavioral outcomes — including EPS enrollment and STI-related out-of-pocket expenditure — to ensure the study assesses real-world financial decision-making rather than abstract knowledge alone.

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A. Platform Design and Technical Specifications

A.1 Technology Stack

Ruta Financiera: El Legado is implemented as a React/TypeScript single-page web application served via a Vite build pipeline. The frontend is styled with Tailwind CSS and designed for full responsiveness across mobile viewport sizes (320–768px). The application requires no installation and is accessible through any modern mobile browser (Chrome, Safari, Firefox, Samsung Internet) via a URL link, allowing distribution through WhatsApp, SMS, or QR code — channels already used extensively by the target population.

The AI conversational components query a large language model API with module-specific system prompts that constrain response style, cultural register, and pedagogical function. Each prompt is engineered to: maintain the guide's persona and narrative voice; respond adaptively to learner input; use Colombian financial referents; generate multimodal content (text + optional image) where pedagogically useful; and emit the `[CORRECTO]` completion token only when substantively correct reasoning is demonstrated. Prompts are versioned and logged to enable reproducibility analysis.

A.2 Engagement and Gamification Architecture

The gamification system is built on a point-gate mechanism: users earn XP through three tiers of interaction (Concept Tutor: 10 XP; Exercise Coach: 20 XP; Real Simulator: 30 XP per interaction), with module completion awarding a 100 XP bonus. Modules 2–5 unlock progressively at thresholds of 150, 350, 600, and 900 XP respectively. Within each module, the Exercise Coach unlocks after the Concept Tutor has generated ≥ 30 XP, and the Simulator unlocks after ≥ 60 total XP is accumulated in the module — ensuring sequential engagement. Threshold values were calibrated in a pilot study with 24 participants in Bogotá.

A.3 Server-Side Data Logging

The platform logs the following events server-side: session start/end timestamps; module entry and exit; chatbot message send/receive; XP award events; module completion events; and any error events. Log entries are timestamped to UTC and include only the participant study ID — no personally identifying information is logged at the platform level. Logs are retained for the duration of the study and three years thereafter, then securely deleted per the data management agreement.

B. Financial Literacy Score: Sample Items

The following are illustrative items from the FLS-27 instrument. The full instrument is available in the Pre-Analysis Plan.

Numeracy: "Si tienes $100,000 en una cuenta de ahorros y la tasa de interés es del 2% anual, después de un año tienes: (a) exactamente $102,000; (b) más de $102,000; (c) menos de $102,000."

Inflation: "Si la inflación es del 5% este año y tu salario sube un 3%, tu capacidad de compra: (a) sube; (b) baja; (c) se mantiene igual."

Datacrédito (Colombian-specific): "En Colombia, si dejas de pagar una deuda durante más de 60 días, cuánto tiempo puede permanecer ese reporte negativo en Datacrédito? (a) 1 año; (b) 2 años; (c) 4 años; (d) 7 años."

EPS incapacity benefit (new item): "Una trabajadora afiliada a una EPS es certificada como incapacitada por 30 días debido a una infección. A partir del día 3, ¿qué porcentaje de su IBC recibe como subsidio de incapacidad? (a) 0%; (b) 50%; (c) 66.7%; (d) 100%."

Risk diversification: "Comprar acciones de una sola empresa es generalmente más riesgoso que comprar un fondo que tenga acciones de muchas empresas. Verdadero o Falso?"

C. Pre-Analysis Plan Registration Template

The PAP will be registered prior to enrollment commencement and will include: (1) study description and hypotheses; (2) exact outcome variable definitions and coding rules; (3) primary and secondary statistical tests with exact regression specifications; (4) all pre-specified subgroup analyses with the interaction terms to be estimated; (5) multiple comparison correction method (FWER for primary, FDR for secondary outcomes); (6) attrition analysis protocol and Lee (2009) bounds procedure; (7) power calculations with all underlying assumptions; (8) data availability and sharing plan; (9) a deviation protocol specifying how any post-hoc changes to the analysis plan will be disclosed in the final manuscript.